

ACADEMENT

AI-Powered Student Companion

Software Requirements Specification (SRS)

Conforming to IEEE Std 830-1998

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for Academent, an AI-powered academic companion platform. The document is intended for software developers, system architects, project managers, and QA engineers. It conforms to IEEE Std 830-1998 and serves as the contractual basis between the development team and all stakeholders.

1.2 Scope

Academent is a cross-platform (web and mobile) AI-powered study companion that helps students organize study materials, ask academic questions, generate quizzes, and track learning progress. It integrates Firebase, LLM APIs (OpenAI / Google Gemini), and native mobile capabilities to deliver a seamless, personalized academic experience.

Primary objectives:

- Provide AI-driven question answering and concept explanations.
- Automate summarization of uploaded academic documents.
- Generate and evaluate personalized quizzes from study materials.
- Track learning progress and identify weak academic areas.
- Support collaborative study through groups, shared resources, and discussion forums.
- Deliver timely reminders and notifications for study sessions and deadlines.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
AI	Artificial Intelligence – LLM-based services (OpenAI GPT / Google Gemini).
SRS	Software Requirements Specification – this document.
FR	Functional Requirement – a requirement specifying system behaviour.
NFR	Non-Functional Requirement – a quality-attribute requirement.
MVP	Minimum Viable Product – earliest releasable version.
Firebase	Google's Backend-as-a-Service platform (auth, database, storage, messaging).
Firestore	Firebase's scalable NoSQL cloud database.
FCM	Firebase Cloud Messaging – push notification service.
API	Application Programming Interface.
MCQ	Multiple-Choice Question.
LLM	Large Language Model.
CF	Cloud Functions – serverless backend logic on Firebase.
UID	User Identifier – unique string for an authenticated user.
RAG	Retrieval-Augmented Generation – AI technique for context-aware responses.
Sprint	Fixed-length Agile iteration, typically two weeks.

1.4 References

- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications.
- Firebase Documentation – <https://firebase.google.com/docs>
- OpenAI API Reference – <https://platform.openai.com/docs>
- Google Gemini API – <https://ai.google.dev/docs>
- React Documentation – <https://react.dev>
- Flutter Documentation – <https://flutter.dev/docs>

1.5 Overview of Document

Section 2 provides an overall description. Section 3 lists functional requirements. Section 4 covers non-functional requirements. Section 5 describes system architecture. Section 6 presents the use case model. Section 7 details the database schema. Section 8 contains the UML class diagram. Section 9 presents sequence diagrams. Section 10 specifies the API design. Section 11 covers security architecture. Sections 12–13 contain the Agile backlog and sprint plan. Section 14 addresses scalability and the future roadmap.

2. Overall Description

2.1 Product Perspective

Academent is a new, self-contained software product built on a serverless, cloud-native architecture. It interfaces with three external system categories: (1) Firebase platform services; (2) third-party LLM APIs; and (3) end-user devices (browsers and mobile phones). It supplements existing LMS platforms as a personal AI study companion.

2.2 Product Vision

"An AI-powered study companion that helps students organize study materials, ask academic questions, generate quizzes, and track learning progress — accessible anywhere, anytime."

2.3 Product Functions

The primary functional areas of Academent are:

1. User Management – registration, authentication, and profile management.
2. AI Study Assistant – natural language Q&A, concept explanations, and homework guidance.
3. Notes Management – upload, organize, search, and tag academic materials.
4. AI Summarization – generate concise and detailed summaries from uploaded documents.
5. Quiz Generation & Evaluation – AI-generated assessments with automatic grading.
6. Study Planning – AI-generated personalized schedules and task management.
7. Progress Tracking – analytics dashboard, weak-area detection, and performance metrics.
8. Reminder System – push notifications for study sessions, deadlines, and examinations.
9. Collaboration – study groups, resource sharing, and discussion forums.

2.4 User Classes and Characteristics

User Class	Description & Characteristics
Student (Primary)	University or school student with basic smartphone/computer literacy. Primary consumer of all features.
System Administrator	Manages system configuration, monitors AI usage, and handles backend operations. Technical background required.

2.5 Operating Environment

- Web Application: React / Next.js – modern browsers (Chrome, Firefox, Safari, Edge).
- Mobile Application: Flutter – iOS 13+ and Android 8.0+.
- Backend: Firebase Cloud Functions (Node.js runtime 18+).
- AI Layer: OpenAI API or Google Gemini API over HTTPS.
- Network: HTTPS (TLS 1.2+) required for all communications.

2.6 Design and Implementation Constraints

- All AI calls must be routed through Firebase Cloud Functions to prevent client-side API key exposure.
- Firestore security rules must enforce strict user-data isolation.
- File uploads limited to PDF and image formats; maximum 20 MB per file.
- System must remain compliant with relevant data privacy regulations.
- Quiz questions must be writable only by Cloud Functions (not by clients directly).

2.7 Assumptions and Dependencies

- Students have access to a stable internet connection.
- Firebase services remain available with current API contracts.
- The chosen LLM API continues to support required prompt formats.
- Development team follows Agile/Scrum with two-week sprints.

3. Specific Requirements – Functional

The table below enumerates all functional requirements. Priority: High = must-have for MVP, Medium = important for release, Low = future enhancement.

3.1 User Management

FR ID	Requirement	Priority
FR-1	Students shall create an account using email/password.	High
FR-1b	Students shall sign up using Google OAuth authentication.	High
FR-2	Students shall log in securely using registered credentials.	High
FR-2b	Students shall reset forgotten passwords via email verification.	High
FR-3	Students shall edit personal information (name, photo, university, degree, year).	Medium
FR-3b	Students shall select academic subjects and interest areas for personalization.	Medium

3.2 AI Study Assistant

FR ID	Requirement	Priority
FR-4	Students shall ask academic questions in natural language through a chat interface.	High
FR-4b	The AI shall generate explanations and answers to submitted questions.	High
FR-5	Students shall request simplified explanations of difficult academic concepts.	High
FR-5b	The AI shall provide explanations calibrated to beginner, intermediate, or advanced levels.	Medium
FR-6	Students shall upload homework questions (text or image) for AI-assisted guidance.	Medium
FR-6b	The AI shall provide hints and step-by-step guidance without directly solving homework.	Medium
FR-7	Students shall upload notes, PDFs, or lecture materials for AI summarization.	High
FR-7b	The AI shall generate concise short summaries, detailed summaries, and bullet-point key extracts.	High
FR-8	Students shall generate quizzes from their uploaded study materials.	High
FR-8b	The AI shall produce multiple-choice and short-answer questions from uploaded content.	High

3.3 Study Planning

FR ID	Requirement	Priority
FR-9	Students shall create named study goals with target completion dates.	Medium
FR-10	The AI shall generate personalized study schedules based on stated goals and available hours.	Medium
FR-10b	The AI shall dynamically adjust study plans based on recorded progress.	Low
FR-11	Students shall create, edit, and delete individual study tasks.	Medium
FR-11b	Students shall mark tasks as completed and view completion history.	Medium

3.4 Progress Tracking

FR ID	Requirement	Priority
FR-12	The system shall automatically track cumulative study hours per user.	High
FR-12b	The system shall track completed tasks, generated quizzes, and attempted assessments.	High
FR-13	Students shall view a dashboard displaying overall progress reports.	High
FR-13b	Students shall view subject-wise performance metrics including average scores.	Medium
FR-14	The AI shall identify weak subjects and topics based on quiz performance.	Medium
FR-14b	The AI shall recommend targeted improvement strategies for weak areas.	Medium

3.5 Smart Notes Management

FR ID	Requirement	Priority
FR-15	Students shall create text-based notes directly within the application.	High
FR-16	Students shall upload PDF files, images, and lecture notes.	High
FR-17	The AI shall automatically categorize uploaded notes by subject using content analysis.	Medium
FR-18	Students shall search their note library using natural language queries.	Medium

3.6 Quiz and Assessment

FR ID	Requirement	Priority
FR-19	Students shall attempt AI-generated quizzes associated with their uploaded notes.	High
FR-20	The system shall automatically evaluate quiz answers and compute scores.	High
FR-21	The AI shall recommend specific topics for revision based on quiz performance.	Medium

3.7 Reminder System

FR ID	Requirement	Priority
FR-22	Students shall receive push notifications for scheduled study sessions.	Medium
FR-23	Students shall receive push notifications for upcoming assignment deadlines.	Medium
FR-24	Students shall receive push notifications for scheduled examinations.	Medium

3.8 Collaboration Features

FR ID	Requirement	Priority
FR-25	Students shall create and join named study groups.	Low
FR-26	Students shall share notes and study materials within study groups.	Low
FR-27	Students shall participate in threaded discussion forums within study groups.	Low

4. Specific Requirements – Non-Functional

NFR ID	Category	Requirement	Metric
NFR-1	Performance	The system shall respond to AI queries within 5 seconds under normal conditions.	≤ 5 s
NFR-2	Performance	The system shall support at least 1,000 simultaneous active users.	$\geq 1,000$
NFR-3	Performance	Page/screen loading time shall not exceed 3 seconds on standard broadband.	≤ 3 s
NFR-4	Reliability	System uptime shall be at least 99% measured monthly.	$\geq 99\%$
NFR-5	Reliability	User data shall be backed up automatically daily with zero data loss.	Daily
NFR-6	Reliability	The system shall recover from transient failures without permanent data loss.	RTO ≤ 30 min
NFR-7	Security	User passwords shall be stored using industry-standard cryptographic hashing.	Mandatory
NFR-8	Security	All communications shall be encrypted using HTTPS/TLS.	TLS 1.2+
NFR-9	Security	Users shall access only their own private data; cross-user access is prohibited.	Mandatory
NFR-10	Security	The system shall comply with applicable data protection regulations.	Mandatory
NFR-11	Usability	The system shall deliver a fully functional experience on both web and mobile.	Both
NFR-12	Usability	First-time users shall be able to use core features without training within 10 minutes.	≤ 10 min
NFR-13	Usability	All core features shall be reachable within three clicks from the home screen.	≤ 3 taps
NFR-14	Scalability	The architecture shall support horizontal scaling without redesign.	Serverless
NFR-15	Scalability	Firebase services shall handle increasing loads up to 10,000 users automatically.	$\geq 10,000$
NFR-16	Maintainability	The system shall be implemented using a modular architecture.	Modular
NFR-17	Maintainability	All source code modules shall include inline documentation.	100% modules
NFR-18	Maintainability	New AI models shall be integrable without changes to the business logic layer.	Pluggable
NFR-19	Availability	The application shall be available 24 hours a day, 7 days a week.	24/7
NFR-20	Availability	Scheduled maintenance windows shall not exceed 2 hours per calendar month.	≤ 2 h/month

5. System Architecture

5.1 Architectural Style

Academent adopts a Client–Server + Serverless architecture. Firebase eliminates custom-managed backend servers. Cloud Functions implement business logic and AI integration middleware. This pattern enables rapid development, automatic scaling, and reduced operational overhead.

5.2 High-Level Architecture Diagram

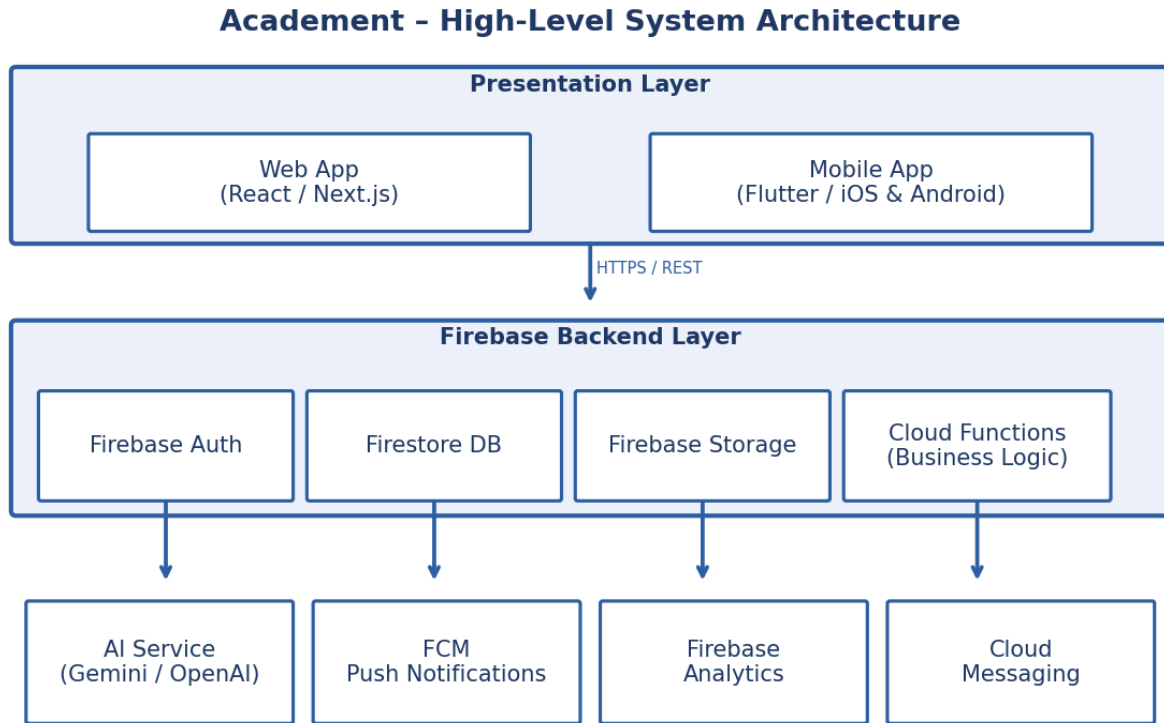


Figure 1 – Academent High-Level System Architecture

5.3 System Context Diagram

C4 System Context Diagram

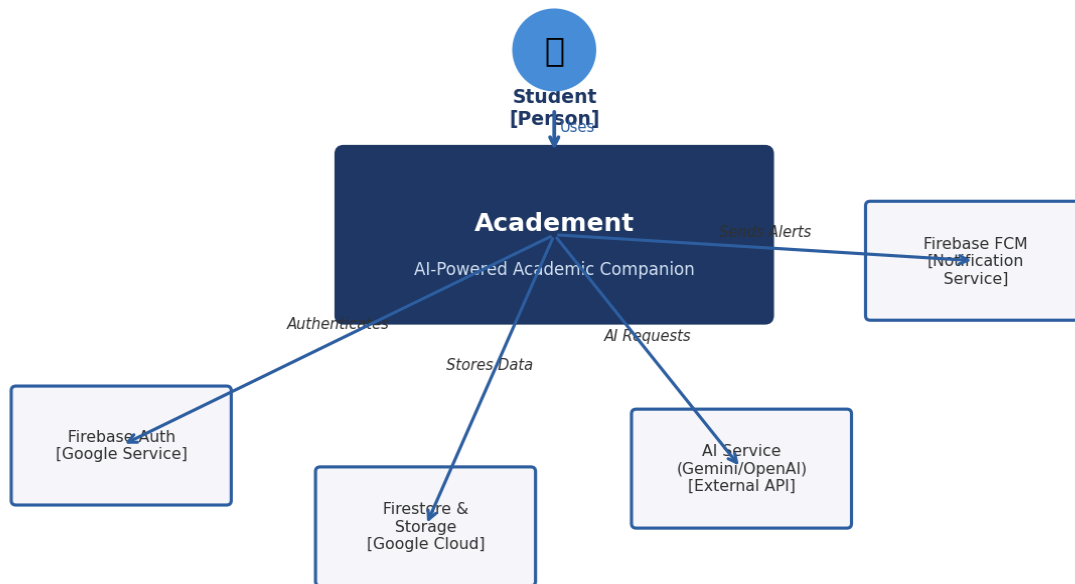
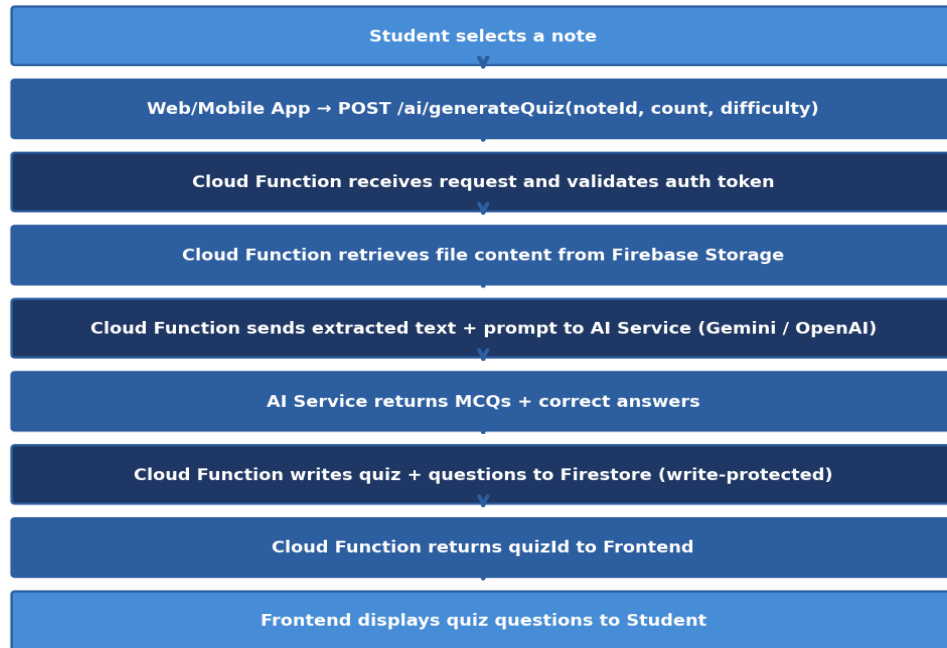
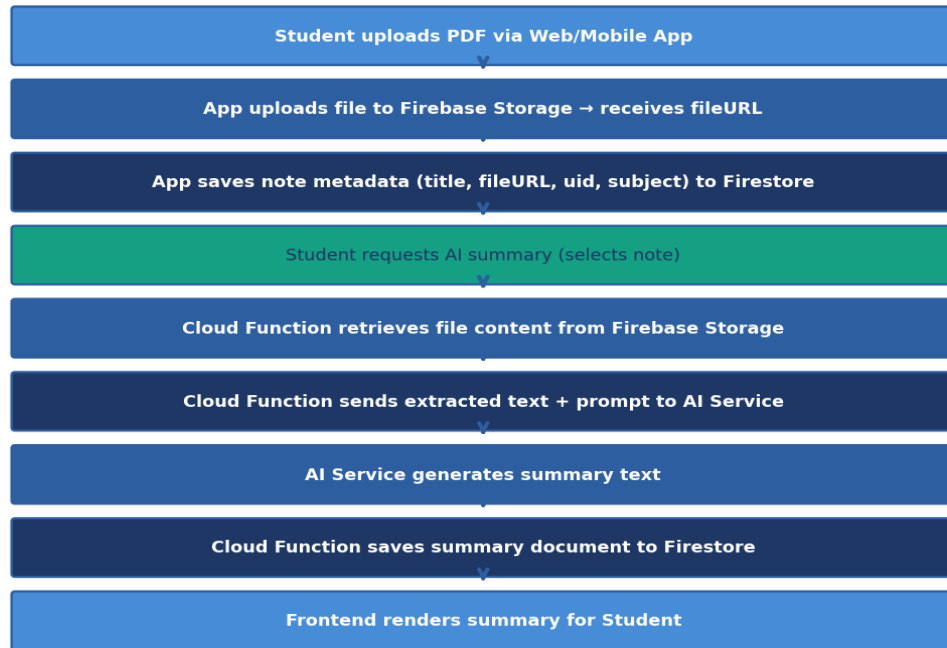


Figure 2 – C4 System Context Diagram

5.4 Data Flow – AI Quiz Generation

Figure 3 - AI Quiz Generation Data Flow**Figure 3 – AI Quiz Generation Data Flow**

5.5 Data Flow – Note Upload and Summarization

Figure 4 – Note Upload & Summarization Data Flow**Figure 4 – Note Upload and Summarization Data Flow**

6. Use Case Model

6.1 Actors

Actor	Description
Student	Primary actor. Interacts with all user-facing system features.
AI Service	External LLM system (Gemini/OpenAI) that processes AI requests.
Firebase Auth	External authentication service that handles credential verification.
Notification Service	Firebase Cloud Messaging that delivers push alerts to devices.

6.2 Use Case Diagram

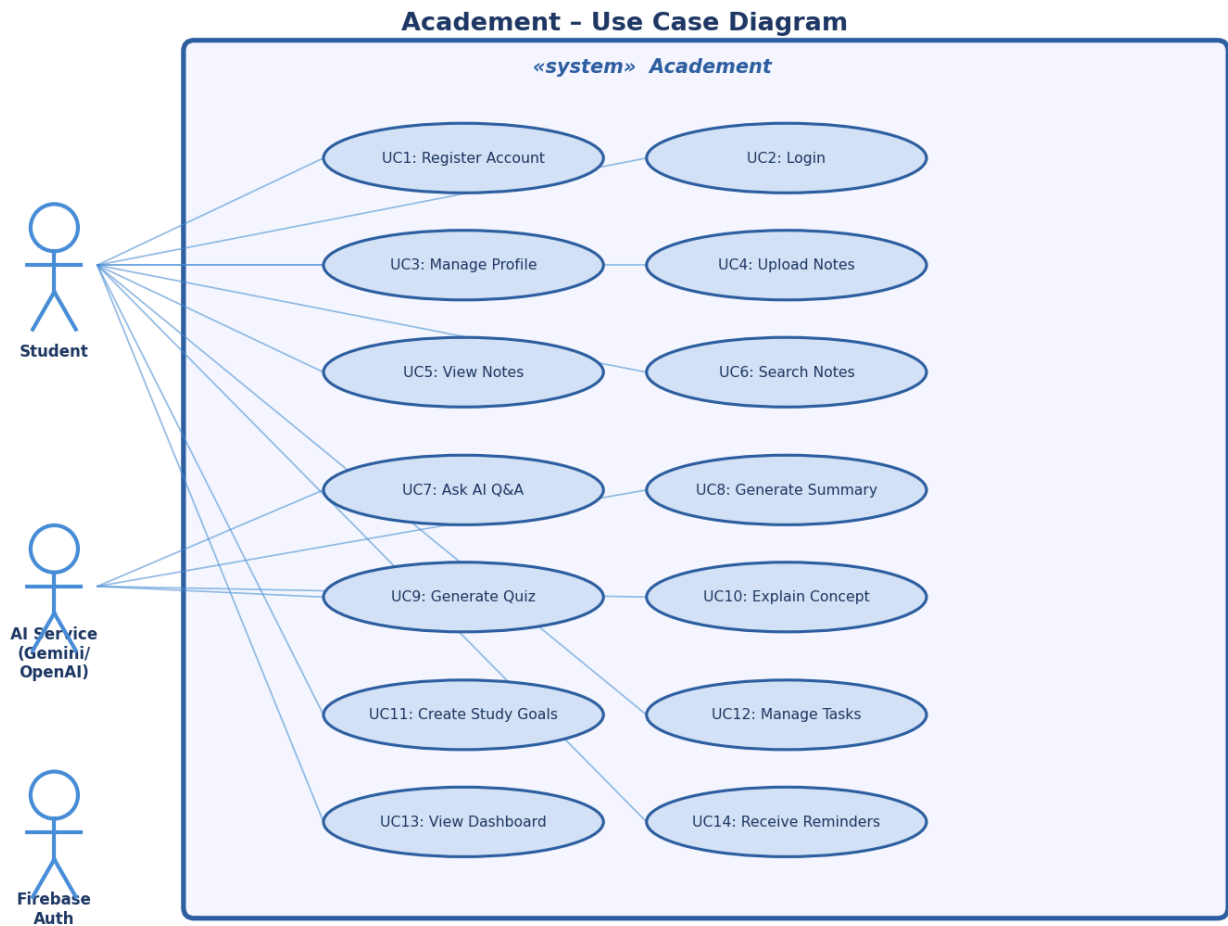


Figure 5 – Academent Use Case Diagram

6.3 Use Case Descriptions

UC ID	Use Case	Description
UC1	Register Account	Student creates an account via email/password or Google OAuth.
UC2	Login	Student authenticates and receives a session token.
UC3	Manage Profile	Student updates personal information, academic details, and preferences.
UC4	Upload Notes	Student uploads PDFs, images, or text notes to Firebase Storage.
UC5	View Notes	Student browses previously uploaded notes with metadata.
UC6	Search Notes	Student queries their note library using natural language.
UC7	Ask Academic Questions	Student submits natural language questions to the AI study assistant.
UC8	Generate Summary	Student requests AI summary of a selected uploaded note.
UC9	Generate Quiz	Student triggers AI quiz generation from a selected note.
UC10	Explain Concept	Student requests a leveled explanation of a concept from AI.
UC11	Create Study Goals	Student defines study goals with target dates.
UC12	Manage Tasks	Student creates, edits, completes, and deletes study tasks.
UC13	View Progress Dashboard	Student views progress analytics: scores, hours, subject performance.
UC14	Receive Reminders	Student receives FCM push notifications for sessions and deadlines.

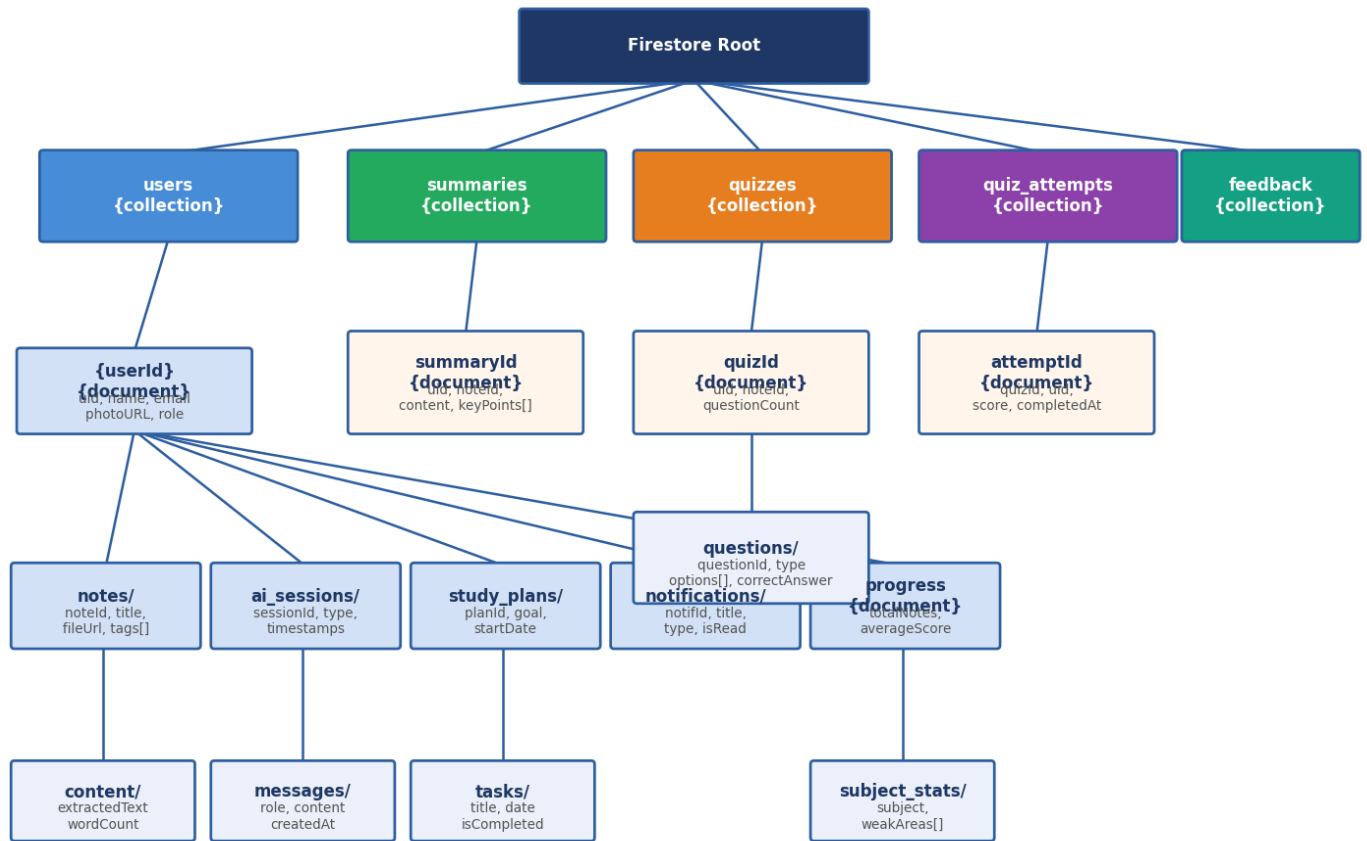
7. Database Schema

Academent uses Google Firebase Firestore, a NoSQL document-oriented cloud database. Data is organised into collections, documents, and subcollections. MVP-critical collections are marked accordingly.

7.1 Collection Overview

Collection	Purpose	Scope	MVP
users/{uid}	Core user document + profile, academic info, preferences.	User-scoped	Yes
users/{uid}/notes	Note metadata for uploaded files.	Subcollection	Yes
users/{uid}/ai_sessions	AI chat sessions with message subcollections.	Subcollection	Yes
users/{uid}/study_plans	Study plans with task subcollections.	Subcollection	Later
users/{uid}/notifications	In-app and push notifications.	Subcollection	Later
users/{uid}/progress	Aggregate learning progress document.	Subcollection	Yes
summaries/{id}	AI-generated note summaries.	Global	Yes
quizzes/{id}	AI-generated quiz metadata and questions.	Global	Yes
quiz_attempts/{id}	Student quiz attempt records and answers.	Global	Yes
feedback/{id}	Student feedback on AI quality and app.	Global	Later

7.2 Firestore Collection Hierarchy Diagram

Figure 6 - Academent Firestore Data Hierarchy**Figure 6 – Academent Firestore Data Hierarchy**

7.3 Key Design Decisions

- User-centric subcollections (notes, ai_sessions, progress) nested under users/{uid} for fast per-user querying and simplified security rule enforcement.
- Global collections (quizzes, summaries, quiz_attempts) separated for cross-user analytics and independent access control.
- This hybrid model represents Firestore best practice for AI-powered applications with strong data isolation requirements.

8. UML Class Diagram

The class diagram presents the full domain model of Academent, covering all subsystems: user management, notes, AI assistant, summarization, quiz, study planning, progress tracking, notifications, and external infrastructure services.

Figure 1. UML Class Diagram

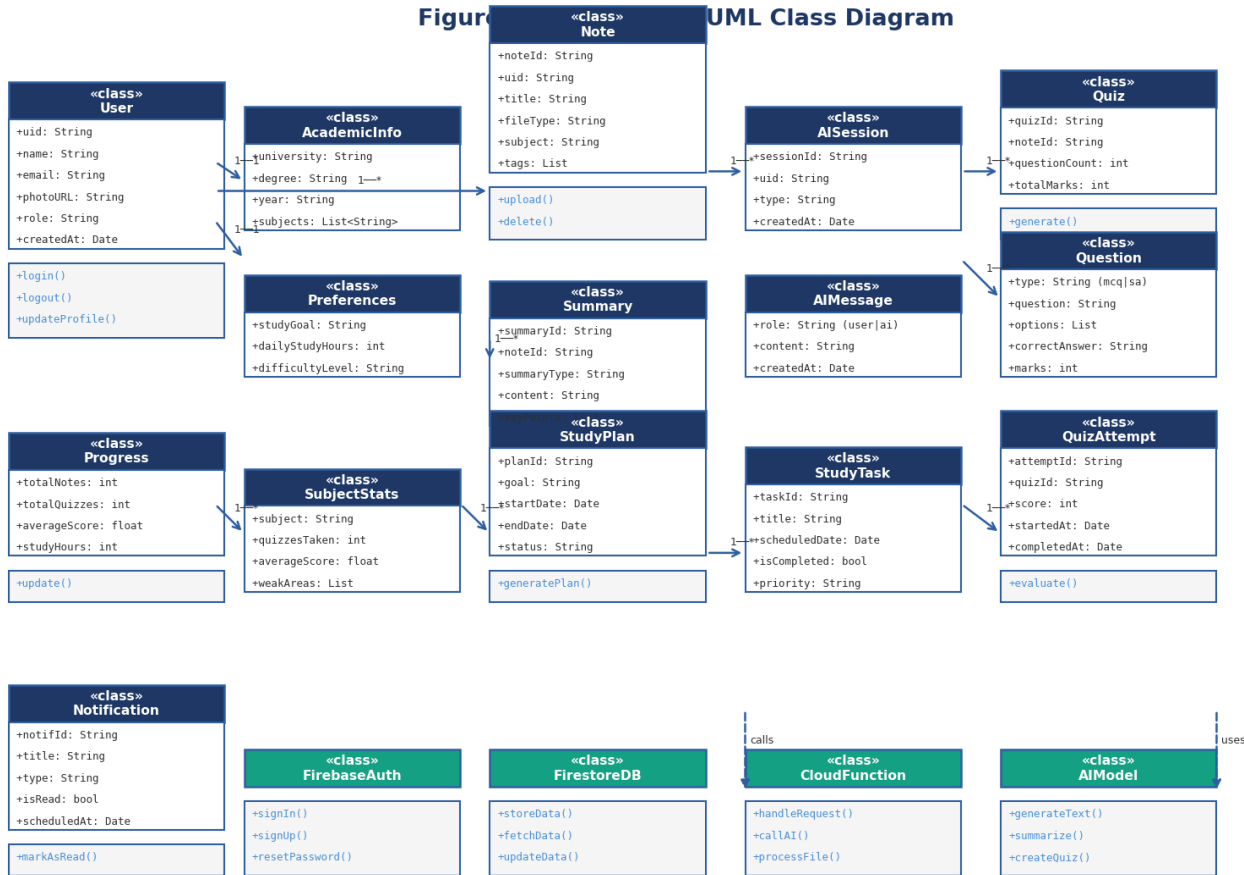


Figure 7 – Academent UML Class Diagram (Full Integrated)

8.1 Key Class Relationships

- User 1—1 AcademicInfo; User 1—1 Preferences; User 1—* Note
- User 1—* AISession; AISession 1—* AIMessage
- Note 1—* Summary; Note 1—* Quiz; Quiz 1—* Question
- User 1—* QuizAttempt; QuizAttempt 1—* Answer
- User 1—1 Progress; Progress 1—* SubjectStats
- User 1—* StudyPlan; StudyPlan 1—* StudyTask
- AISession → CloudFunction → AIModel (dependency chain)

9. Sequence Diagrams

The following sequence diagrams describe the interaction flows for all major system workflows. Solid arrows represent requests/calls; dashed arrows represent returns/responses.

9.1 User Registration Flow

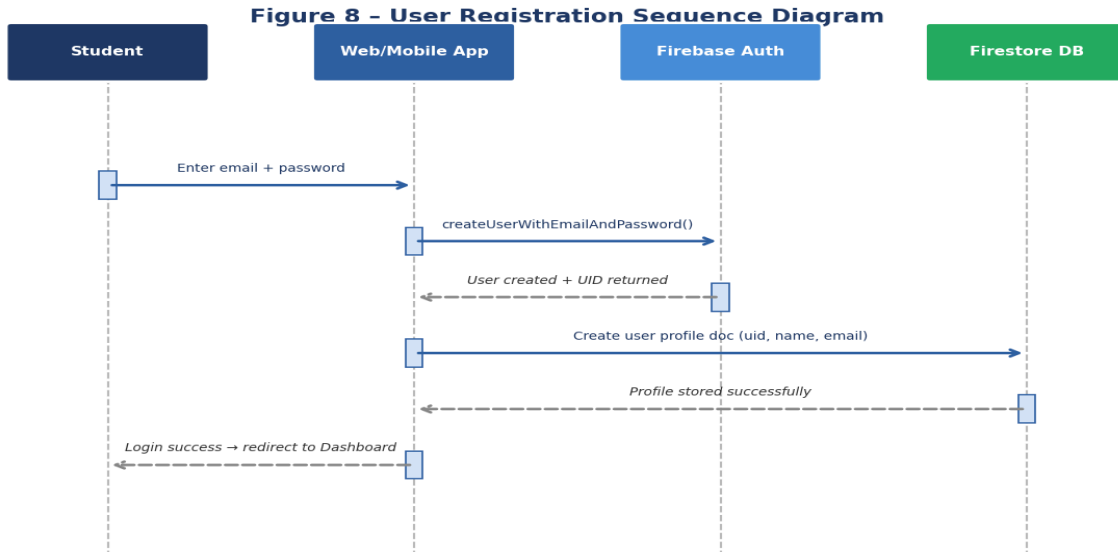


Figure 8 – User Registration Sequence Diagram

9.2 User Login Flow

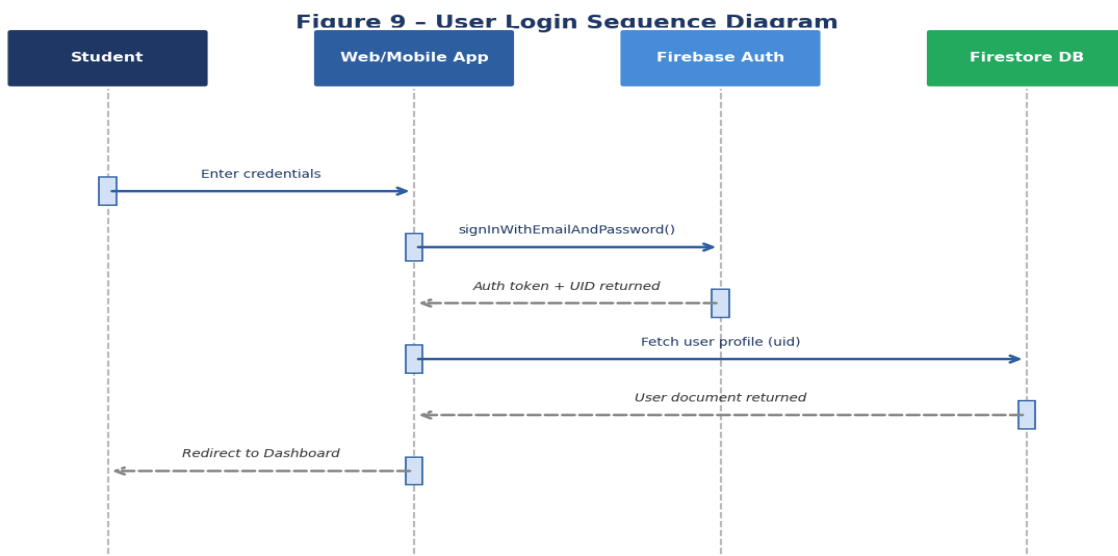


Figure 9 – User Login Sequence Diagram

9.3 Upload Notes Flow

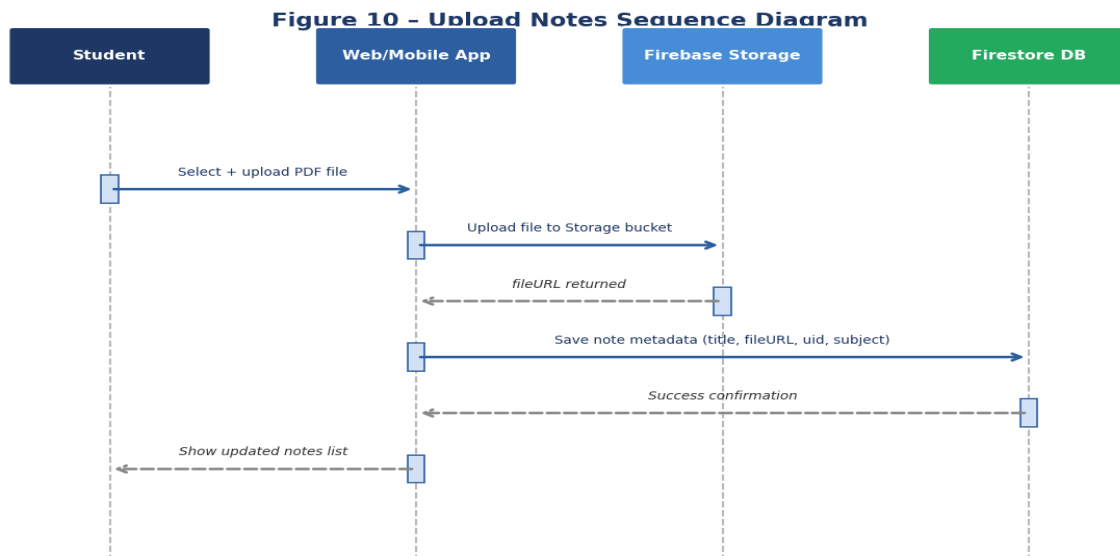


Figure 10 – Upload Notes Sequence Diagram

9.4 AI Question Answering Flow

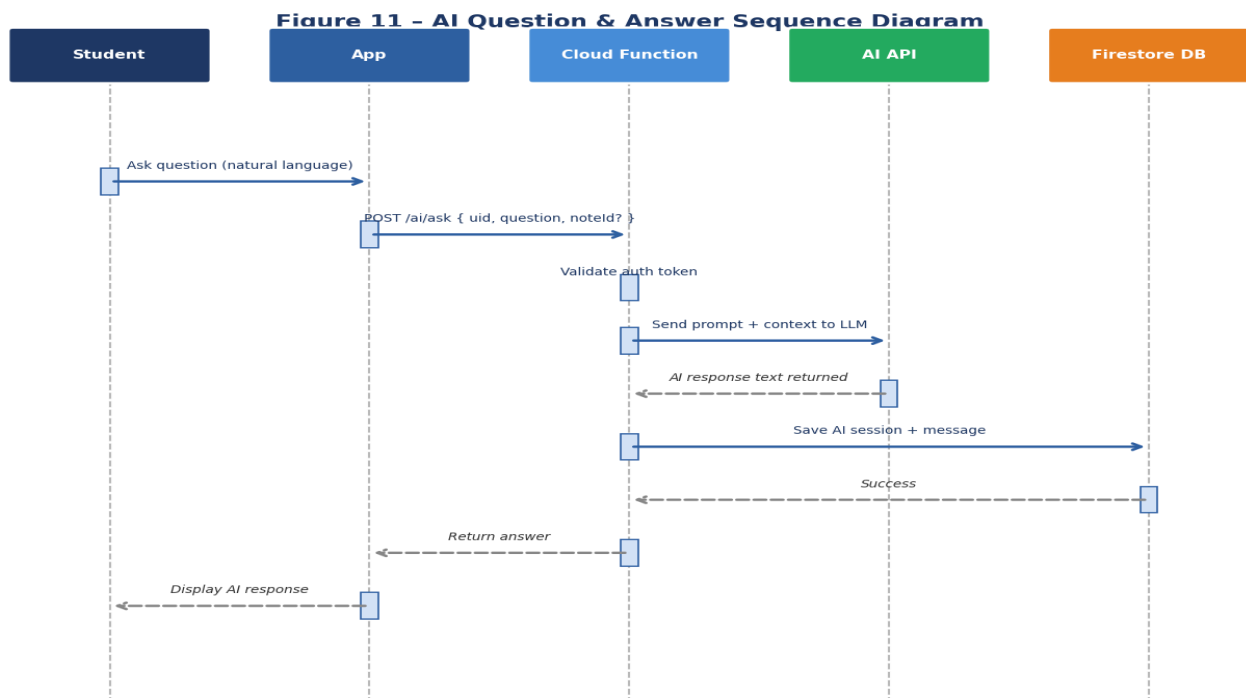


Figure 11 – AI Question Answering Sequence Diagram

9.5 AI Note Summarization Flow

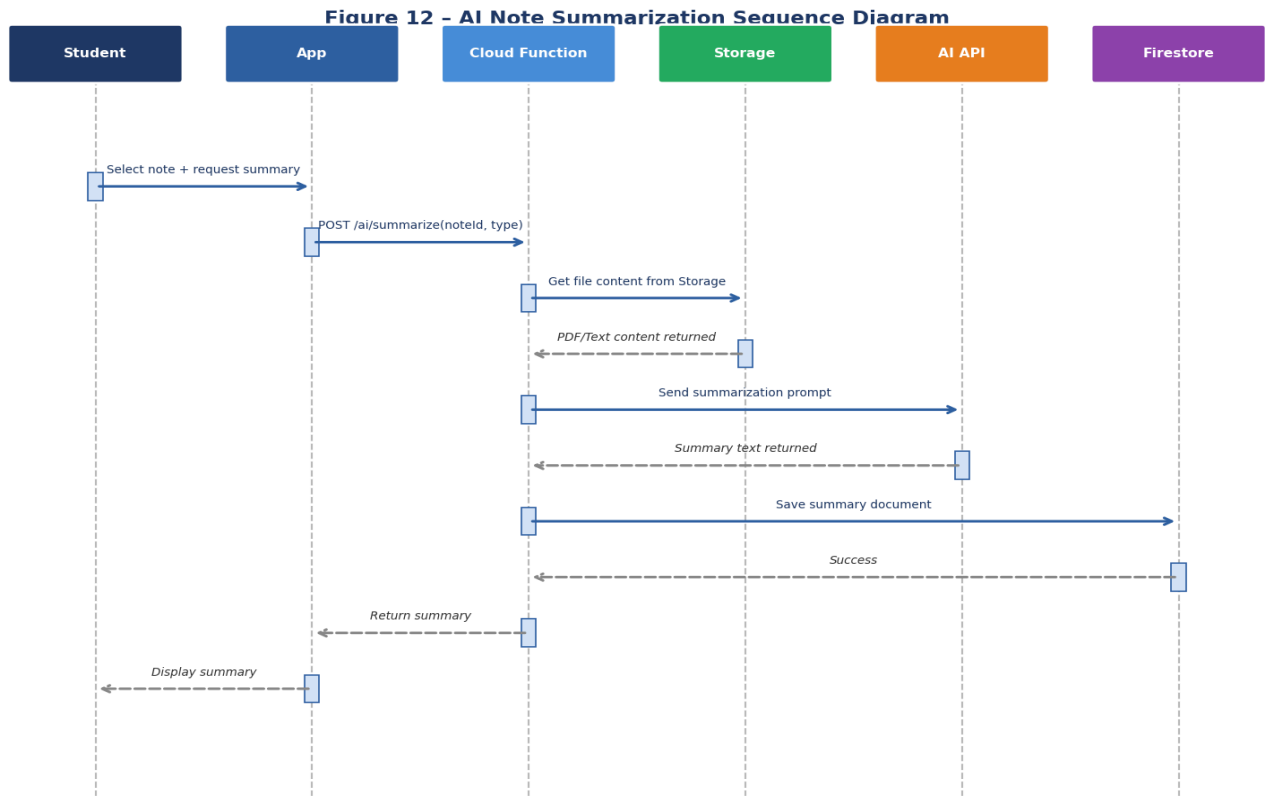


Figure 12 – AI Note Summarization Sequence Diagram

9.6 Quiz Generation Flow

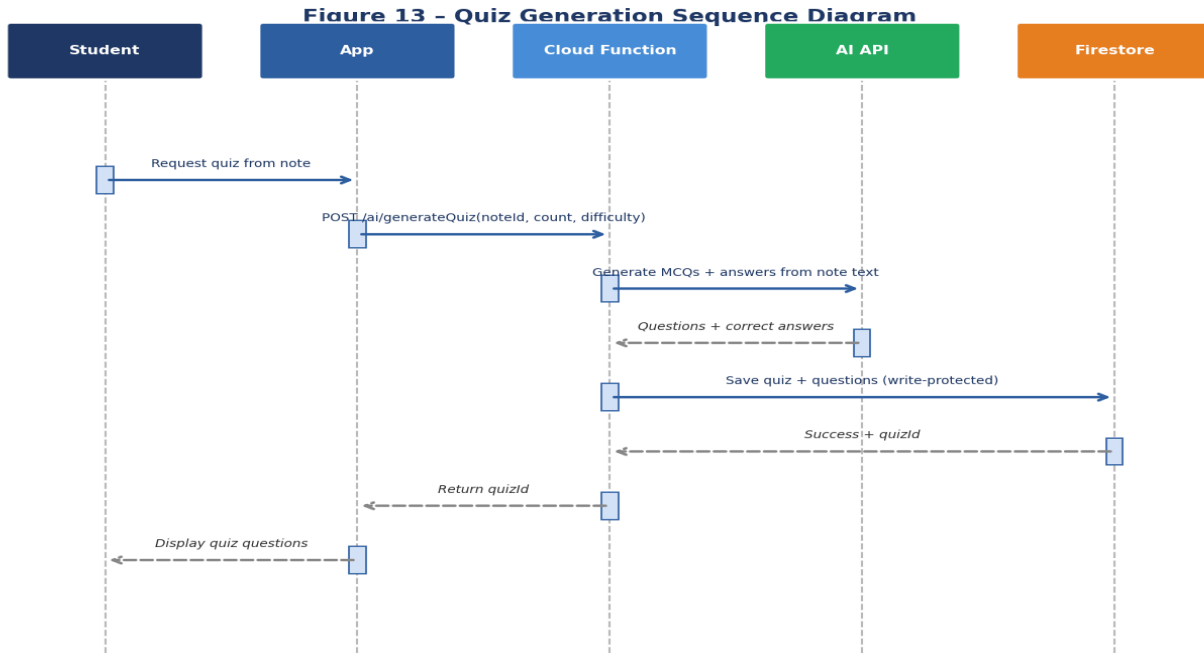


Figure 13 – Quiz Generation Sequence Diagram

9.7 Quiz Submission and Evaluation Flow

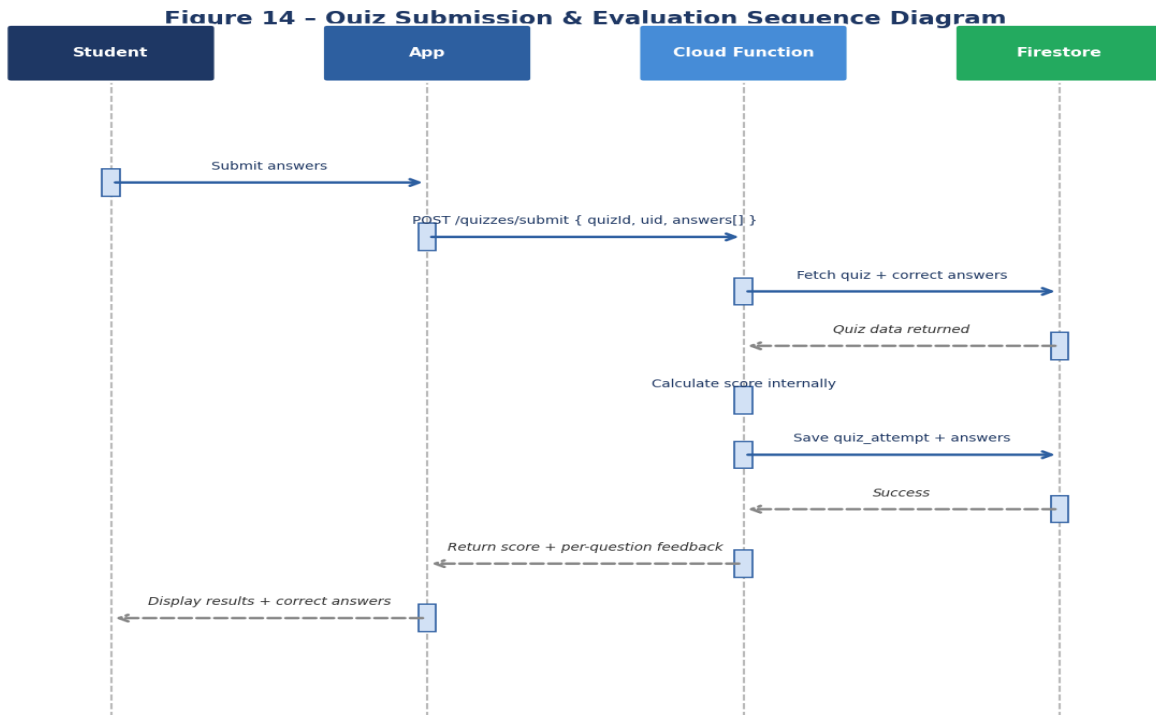


Figure 14 – Quiz Submission and Evaluation Sequence Diagram

9.8 Study Plan Generation Flow

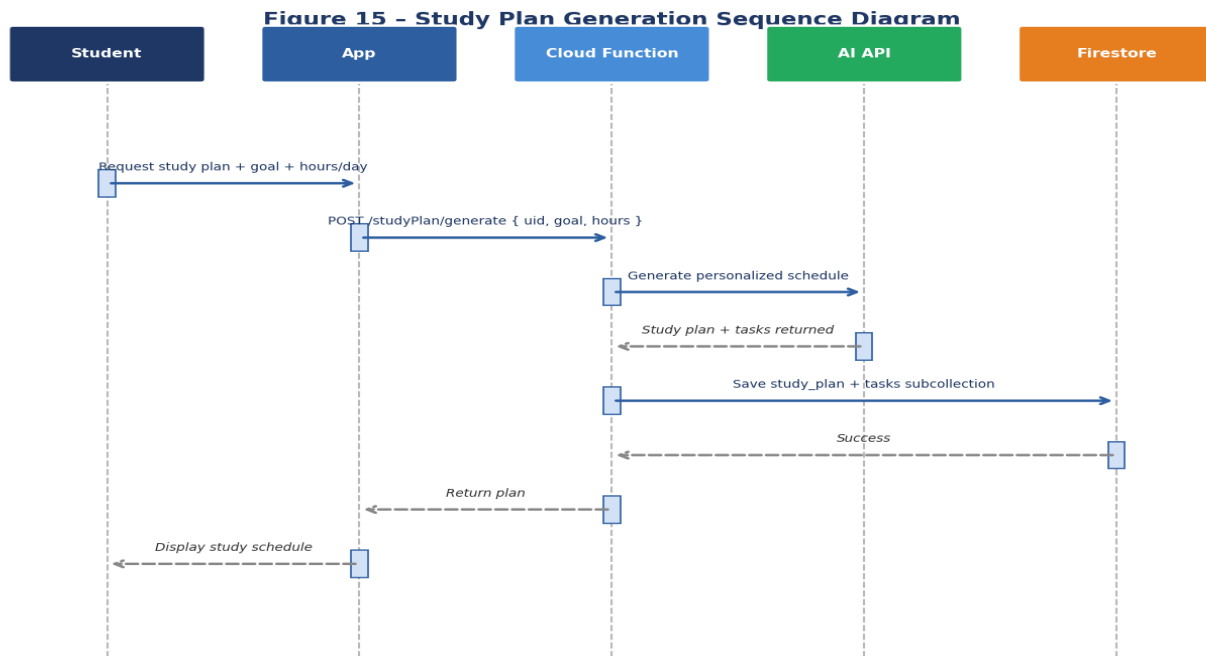


Figure 15 – Study Plan Generation Sequence Diagram

9.9 Progress Tracking Flow

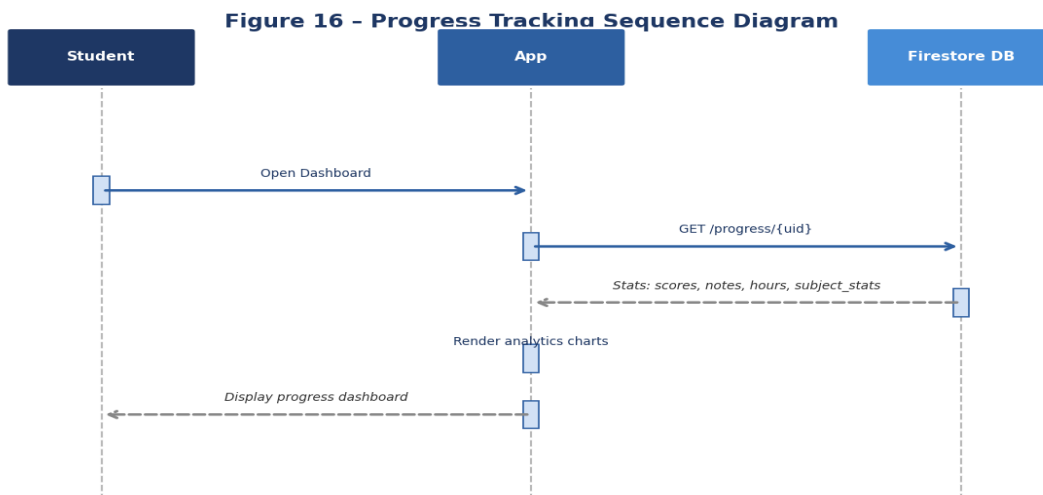


Figure 16 – Progress Tracking Sequence Diagram

9.10 Notification Delivery Flow

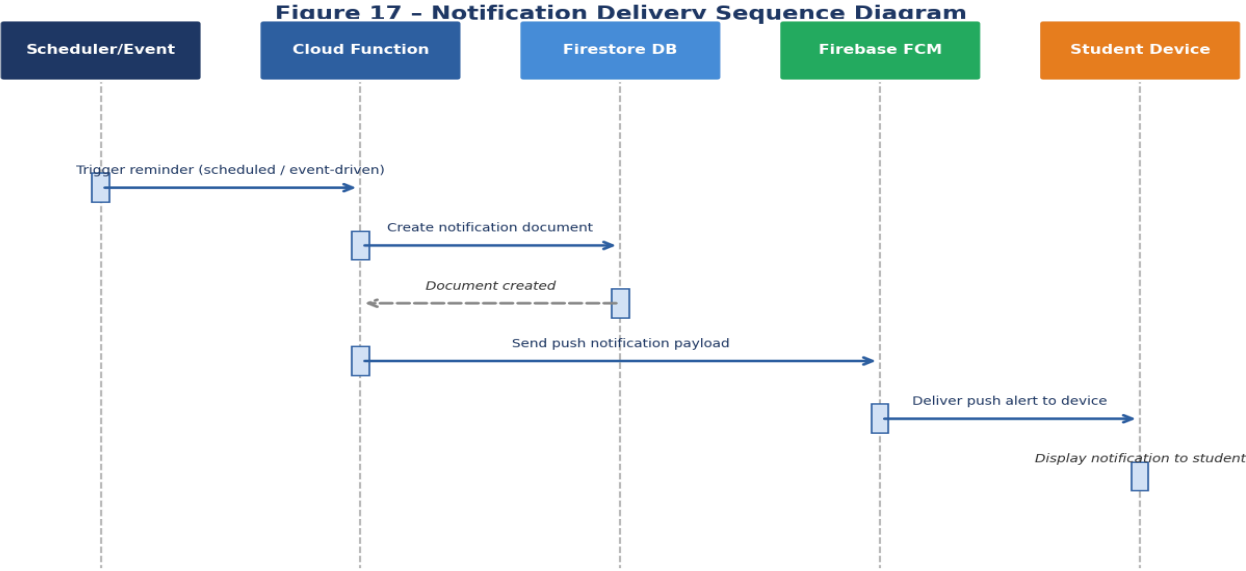


Figure 17 – Notification Delivery Sequence Diagram

10. API Design

All AI and business logic is exposed through Firebase Cloud Functions REST-style endpoints. The base URL is: <https://<region>-<project-id>.cloudfunctions.net/api>. All endpoints require a valid Firebase Authentication bearer token. No AI API keys are ever exposed to client-side code.

10.1 User Management Endpoints

Endpoint	Method	Description
/users/createProfile	POST	Create Firestore user document after Firebase Auth signup.
/users/{uid}	GET	Retrieve user profile document.
/users/{uid}	PUT	Update user profile fields.
/users/{uid}	DELETE	Delete user account and associated data.

10.2 Notes Endpoints

Endpoint	Method	Description
/notes	POST	Save note metadata after file upload to Firebase Storage.
/notes?uid={uid}	GET	Retrieve all notes belonging to a user.
/notes/{noteId}	GET	Retrieve a specific note by ID.
/notes/{noteId}	DELETE	Delete a note document and its Storage file.
/notes/extractText	POST	Extract text from an uploaded PDF for AI processing.

10.3 AI Assistant Endpoints

Endpoint	Method	Description
/ai/ask	POST	Submit a natural language question; returns AI-generated answer and session ID.
/ai/explain	POST	Request a concept explanation at a specified difficulty level.
/ai/summarize	POST	Generate a note summary (short, detailed, or bullet-point format).
/ai/generateQuiz	POST	Generate a quiz from a specified note with configurable count and difficulty.
/ai/session	POST	Create a new AI chat session.
/ai/session?uid={uid}	GET	List all AI sessions for a user.
/ai/session/{sid}/messages	GET	Retrieve all messages in a session.
/ai/session/{sid}/messages	POST	Send a message within a session.

10.4 Quiz Endpoints

Endpoint	Method	Description
/quizzes/{quizId}	GET	Retrieve a quiz by ID.
/quizzes/generate	POST	Trigger AI quiz generation from a note.
/quizzes/submit	POST	Submit quiz answers; returns score and per-question feedback.
/quizzes/results/{attemptId}	GET	Retrieve results of a specific quiz attempt.
/quizzes/history?uid={uid}	GET	Retrieve all quiz attempts by a user.

10.5 Study Plan Endpoints

Endpoint	Method	Description
/studyPlan/generate	POST	AI-generate a personalised study plan based on goal and available hours.
/studyPlan?uid={uid}	GET	Retrieve all study plans for a user.
/studyPlan/task/{taskId}	PUT	Update task completion status.

10.6 Progress & Analytics Endpoints

Endpoint	Method	Description
/progress/{uid}	GET	Retrieve progress dashboard data for a user.
/progress/update	POST	Internal endpoint to update progress after a quiz attempt.
/progress/{uid}/subjects	GET	Retrieve subject-level performance statistics.

10.7 Notification Endpoints

Endpoint	Method	Description
/notifications/create	POST	System-triggered notification creation.
/notifications?uid={uid}	GET	List all notifications for a user.
/notifications/{id}/read	PUT	Mark a notification as read.

10.8 MVP API Priority

Endpoints mandatory for Sprint 1–3 MVP delivery:

- **POST** /users/createProfile
- **POST** /notes and **GET** /notes?uid=
- **POST** /ai/summarize
- **POST** /ai/generateQuiz
- **POST** /ai/ask

- POST /quizzes/submit
- GET /progress/{uid}

11. Security Architecture

11.1 Authentication

All user sessions are managed by Firebase Authentication. Every API call to a Cloud Function must include a valid Firebase Auth ID token as a bearer token in the Authorization header. Tokens are verified server-side before any business logic executes.

11.2 Data Isolation – Firestore Security Rules

Firestore security rules enforce strict per-user data isolation. Users can only read or write documents where the uid field matches their authenticated UID. A default-deny rule blocks all uncovered paths.

```
rules_version = "2"; service cloud.firestore {
  match /users/{userId} {
    allow read, update, delete: if request.auth.uid == userId;
    match /notes/{noteId} { allow read, write: if request.auth.uid == userId; }
  } match /quizzes/{quizId} {
    match /questions/{q} { allow read: if true; allow write: if false; }
  } match /{document=**} { allow read, write: if false; } }
```

11.3 Security Rules – Collection Strategy

Collection	Rule Type	Reason	Notes
summaries	user-scoped read/write	AI outputs belong to the generating user.	uid must match auth
quizzes	user-scoped read/write	Quiz ownership and integrity.	uid must match auth
quiz_attempts	user-scoped	Exam attempt integrity.	uid must match auth
quizzes/questions	read-only for client	Prevent client-side answer exposure.	write: if false
feedback	user-scoped	Privacy of feedback data.	uid must match auth
users subcollections	strict ownership	Core personal data security.	isOwner() function

11.4 AI Key Security

LLM API keys (OpenAI / Gemini) are stored as Firebase Cloud Functions environment secrets and are never transmitted to client applications. All AI requests are proxied through Cloud Functions, ensuring API credentials remain server-side at all times.

12. Agile Product Backlog

Academent is developed using Agile/Scrum with two-week sprints. The product backlog is organized by epic. Priority: High = MVP must-have, Medium = Release 2, Low = future enhancement.

12.1 Epic 1 – User Management

ID	User Story	Priority
UM-01	As a student, I want to register using email and password.	High
UM-02	As a student, I want to sign in using Google authentication.	Medium
UM-03	As a student, I want to edit my profile details.	Medium
UM-04	As a student, I want to delete my account.	Low

12.2 Epic 2 – Notes Management

ID	User Story	Priority
NM-01	As a student, I want to upload lecture notes in PDF format.	High
NM-02	As a student, I want to upload images of notes.	Medium
NM-03	As a student, I want to create text notes manually.	Medium
NM-04	As a student, I want to view my uploaded notes.	High
NM-05	As a student, I want to delete notes I no longer need.	Medium
NM-06	As a student, I want to search my notes using natural language.	Medium

12.3 Epic 3 – AI Academic Assistant

ID	User Story	Priority
AI-01	As a student, I want to ask academic questions in natural language.	High
AI-02	As a student, I want AI to explain difficult concepts.	High
AI-03	As a student, I want AI to provide examples related to concepts.	Medium
AI-04	As a student, I want AI to answer questions based on my uploaded notes.	High
AI-05	As a student, I want AI to explain concepts at different difficulty levels.	Medium

12.4 Epic 4 – AI Summarization

ID	User Story	Priority
AS-01	As a student, I want AI to summarize uploaded notes.	High
AS-02	As a student, I want short summaries for quick revision.	Medium
AS-03	As a student, I want detailed summaries for in-depth study.	Medium
AS-04	As a student, I want key points extracted from documents.	High

12.5 Epic 5 – Quiz Generation

ID	User Story	Priority
QZ-01	As a student, I want AI to generate quizzes from my notes.	High
QZ-02	As a student, I want multiple-choice questions.	High
QZ-03	As a student, I want short-answer questions.	Medium
QZ-04	As a student, I want automatic quiz grading.	High
QZ-05	As a student, I want to review correct answers after a quiz.	High

12.6 Epic 6 – Study Planning

ID	User Story	Priority
SP-01	As a student, I want to create study goals.	Medium
SP-02	As a student, I want to create study tasks.	Medium
SP-03	As a student, I want to mark tasks as completed.	Medium
SP-04	As a student, I want AI-generated study plans.	Medium
SP-05	As a student, I want study schedules based on my deadlines.	Low

12.7 Epic 7 – Progress Tracking

ID	User Story	Priority
PT-01	As a student, I want to view my overall progress dashboard.	High
PT-02	As a student, I want to see subject-wise performance statistics.	Medium
PT-03	As a student, I want AI to identify my weak topics.	Medium
PT-04	As a student, I want to see my quiz history.	Medium

12.8 Epic 8 – Reminder System

ID	User Story	Priority
RM-01	As a student, I want push notifications for study sessions.	Medium
RM-02	As a student, I want notifications for assignment deadlines.	Medium
RM-03	As a student, I want exam reminders.	Medium

12.9 Epic 9 – Collaboration

ID	User Story	Priority
CL-01	As a student, I want to create and join study groups.	Low
CL-02	As a student, I want to share notes within study groups.	Low
CL-03	As a student, I want to participate in group discussion forums.	Low

13. Sprint Plan

13.1 Sprint 1 – Authentication and Notes Foundation

Goal: Build a working application where a student can register, log in, create a profile, upload notes, and view uploaded notes.

Layer	Task	Stories
Frontend	Build Login Page with email/password form and Google sign-in.	UM-01, UM-02
Frontend	Build Registration Page with validation.	UM-01
Frontend	Build Dashboard skeleton page.	PT-01
Frontend	Build Notes Upload Page with file picker.	NM-01, NM-02
Backend	Configure Firebase Authentication (email + Google OAuth).	UM-01, UM-02
Backend	Configure Firestore: users and notes collections.	UM-01
Backend	Configure Firebase Storage for PDF/image uploads.	NM-01, NM-02
Backend	Implement POST /users/createProfile Cloud Function.	UM-01
Backend	Implement POST /notes and GET /notes Cloud Functions.	NM-01, NM-04

13.2 Sprint 2 – AI Core Features

Goal: Integrate AI Q&A, note summarization, and quiz generation.

Feature	Stories	Acceptance Criteria
AI Q&A Chat Interface	AI-01, AI-04	Student can type a question and receive an AI answer within 5 seconds.
Note Summarization	AS-01, AS-04	Student selects a note; AI generates summary in ≤ 15 seconds.
Quiz Generation	QZ-01, QZ-02	Student generates ≥ 5 MCQs from a selected note.
Quiz Evaluation	QZ-04, QZ-05	System auto-grades quiz and shows correct answers.

13.3 Sprint 3 – Progress and Analytics

Goal: Build progress tracking dashboard and begin personalized study planning.

Feature	Stories	Acceptance Criteria
Progress Dashboard	PT-01, PT-02	Dashboard loads in ≤ 3 s; shows notes count, quiz count, avg score.
Subject Performance	PT-02, PT-03	Per-subject quiz averages displayed; weak areas highlighted.
Study Goals + Tasks	SP-01, SP-02, SP-03	Student can create goals, add tasks, and mark tasks complete.

Feature	Stories	Acceptance Criteria
AI Study Plan	SP-04	AI generates a schedule based on user goal and available hours.

13.4 Definition of Done

A feature is considered complete when all of the following criteria are satisfied:

10. All acceptance criteria for the feature are met and verified.
11. Unit tests pass with no critical failures.
12. Integration tests pass in the staging environment.
13. No unresolved high-severity bugs remain.
14. Code has been reviewed and approved by at least one other team member.
15. Feature has been deployed to the testing environment.
16. Product Owner has accepted the feature demonstration.
17. Relevant documentation (in-code and wiki) has been updated.

14. Scalability and Future Roadmap

14.1 Scalability Strategy

- Firebase auto-scales Firestore, Storage, and Cloud Functions independently with zero manual intervention.
- Serverless Cloud Functions scale horizontally under load and reset to zero when idle.
- AI service scaling is delegated to the external LLM provider (OpenAI / Gemini).
- Future: a Redis caching layer may be introduced to reduce repeated LLM API calls.
- Future: a vector database (e.g., Pinecone) may be added for Retrieval-Augmented Generation (RAG) note search.

14.2 Post-MVP Feature Roadmap

Phase	Feature	Technology
Phase 4	Push notification reminders and exam alerts.	Firestore FCM
Phase 4	Study group creation and membership.	Firestore
Phase 5	Resource sharing and discussion forums within groups.	Firestore, Storage
Phase 5	AI-powered weak area detection and recommendations.	Firestore analytics + LLM
Phase 6	Retrieval-Augmented Generation (RAG) note search.	Vector DB (Pinecone)
Phase 6	Adaptive AI tutoring that adjusts to student level.	LLM with memory
Phase 7	Offline support for mobile app.	Flutter Hive / SQLite
Phase 7	Third-party LMS integration (Moodle, Canvas).	REST API connectors

Appendix A – Technology Stack Summary

Layer	Technology	Purpose
Web Frontend	React / Next.js	Server-side rendering, routing, and component UI.
Mobile Frontend	Flutter (Dart)	Cross-platform iOS and Android application.
Authentication	Firebase Authentication	Email/password and Google OAuth user auth.
Database	Firebase Firestore	NoSQL document store for all app data.
File Storage	Firebase Storage	PDF, image, and document file hosting.
Backend Logic	Cloud Functions (Node.js)	Serverless API and AI integration layer.
AI / LLM	OpenAI GPT-4 / Google Gemini	Q&A, summarization, quiz generation, tutoring.
Push Notifications	Firebase Cloud Messaging	Reminders, deadline alerts, notifications.
Analytics	Firebase + Cloud Functions	Progress tracking and subject performance.
Security	Firebase Security Rules + HTTPS	Data isolation and encrypted transmission.

Appendix B – Document Revision History

Version	Date	Author	Changes
1.0	June 2026	Development Team	Initial release – full SRS document with visual diagrams.